

Designing for the Sun: Solar Energy and Efficiency

Energy3D · Concord Consortium

Senior Secondary (Years 11-12)

~30 minutes

Science · Earth & Environmental
Science

NOTICE — AI-generated worksheet

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Simulation: <http://energy.concord.org/energy3d/>

Curriculum links: energy · energy transfer · Earth resources · Earth geometry

Learning intentions

- I can design a simple structure and add solar panels using CAD-style modelling.
- I can use sun-path data to explain how the sun's position changes across the day and across seasons.
- I can evaluate and improve a design's energy output based on simulation results.

Getting started

Open Energy3D and start a new project. Build a simple structure and add at least one solar panel to its roof or a nearby ground-mounted array, then run the built-in sun-path/energy analysis, referring to your Earth & Environmental Science (Unit 3) knowledge of how Earth's tilt affects sunlight across the year.

Tasks

1. Build a basic structure and add one solar panel at a fixed angle. Run the energy analysis for a single day and record the total energy output shown.

2. Change the date setting to a different season. Predict whether the energy output will increase or decrease, then run the analysis and check your prediction.

3. Test the same panel at three different tilt angles (e.g. flat, 30 degrees, 60 degrees). Record the daily energy output for each.

Panel tilt angle	Daily energy output

4. Using your table, explain why panel angle affects how much energy is captured, referring to the sun's path across the sky.

5. Move your solar panel to a different position on the structure (e.g. a different side of the roof) and compare its output to your first result.

6. Explain, using the sun-path model, why the time of year changes the best tilt angle for a solar panel at a fixed location.

7. If your best-performing design produced a certain amount of energy per day in the simulation, estimate how much energy it would produce over a full year, and note one assumption you had to make.

Working:

8. Evaluate your final design: what change made the biggest improvement to energy output, and what trade-off (e.g. cost, shading, structural stability) might limit how far you could push that change in a real building?

Extension (optional)

Real solar farms sometimes use tracking systems that rotate panels to follow the sun through the day. Use your understanding of the sun-path model to explain why a tracking system would produce more energy than a fixed panel.
